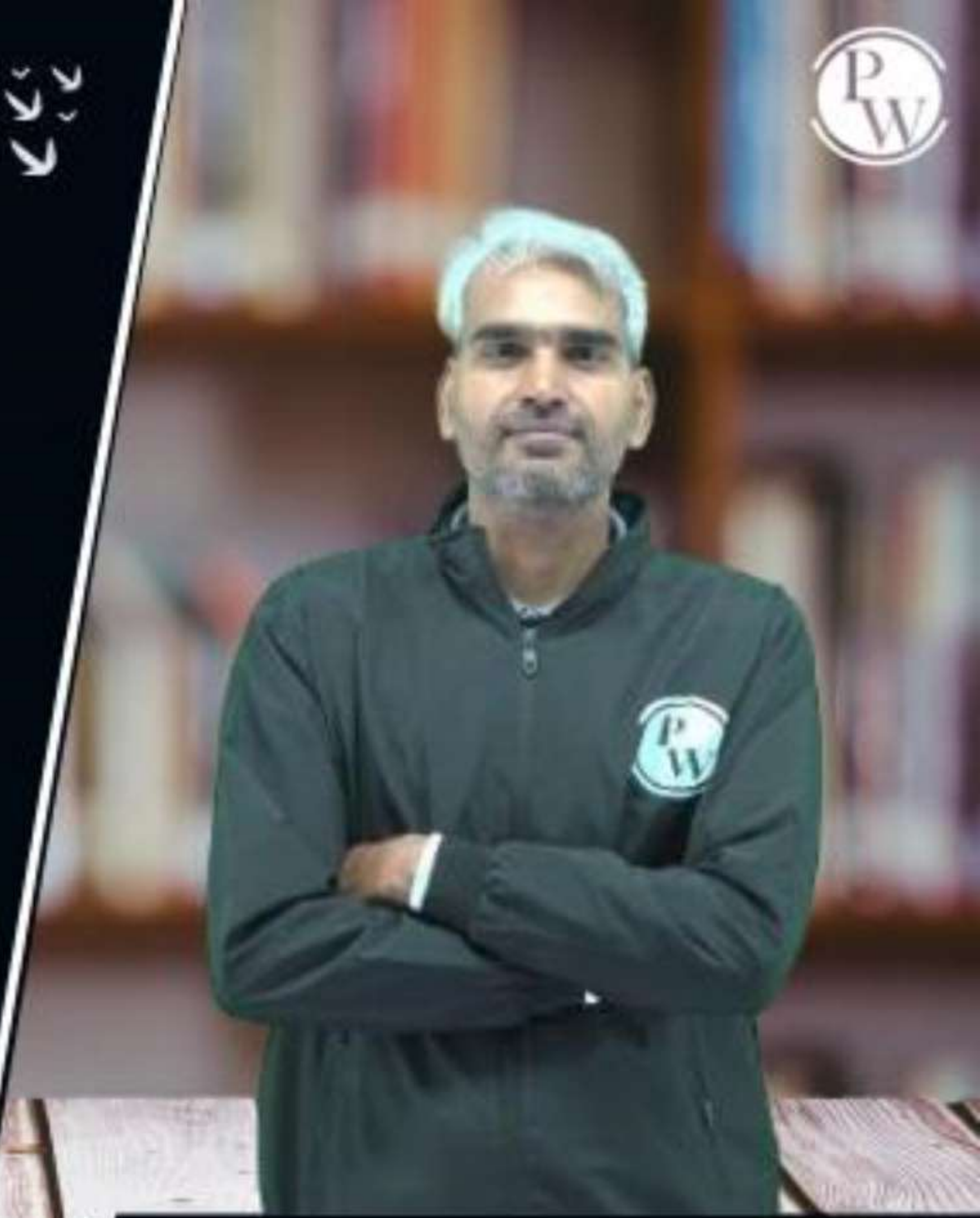
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 04



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Recap of Previous Lecture

Topic Set operations and properties of set operations

Topic Multi-set

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Topics to be Covered

Relation:

Topic	Cartesian product
Topic	Relation
Topic	Different types of relations
Topic	Diagonal relation
Topic	Reflexive relation
Topic	Irreflexive relation

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Topic : Cartesian Product

Cross Product :-

Let A and B are any two sets, Cartesian product of set A and set B is denoted by 'A x B', and it is defined as.

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

ordered pair :→ ∴ The order in which elements appear is important

$\boxed{\text{in general, } (a, b) \neq (b, a)}$

$\boxed{(a,b) = (b,a) \text{ if and only if } a = b}$

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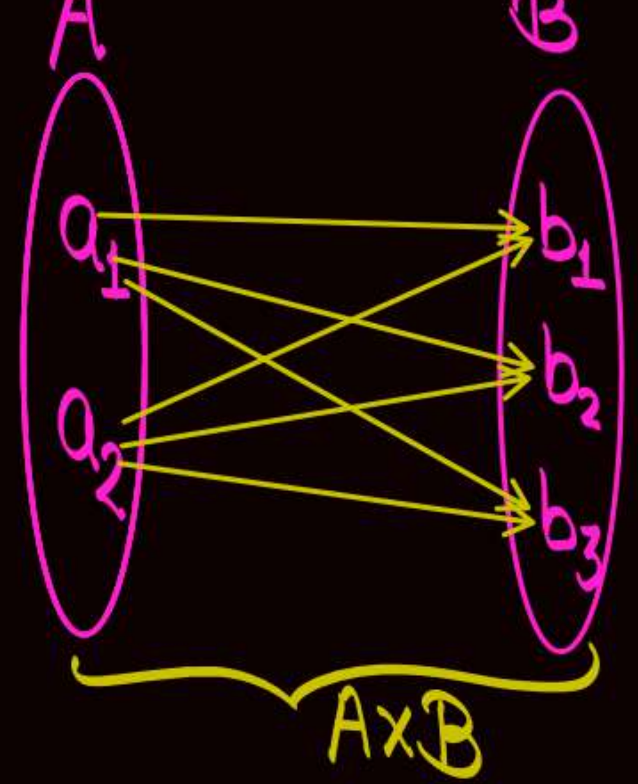
eg:- $A = \{a_1, a_2\}$

$B = \{b_1, b_2, b_3\}$

$A \times B = \{(a_1, b_1), (a_1, b_2), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_2, b_3)\}$



$B \times A = \{(b_1, a_1), (b_1, a_2), (b_2, a_1), (b_2, a_2), (b_3, a_1), (b_3, a_2)\}$

We know, $(a_1, b_1) \neq (b_1, a_1)$

$\therefore$ in general, $A \times B \neq B \times A$

* In general, $A \times B \neq B \times A$.

* $A \times B = B \times A$ if and only if (i) $A = B$

  or (ii) If at least one of A or B is an Empty set.

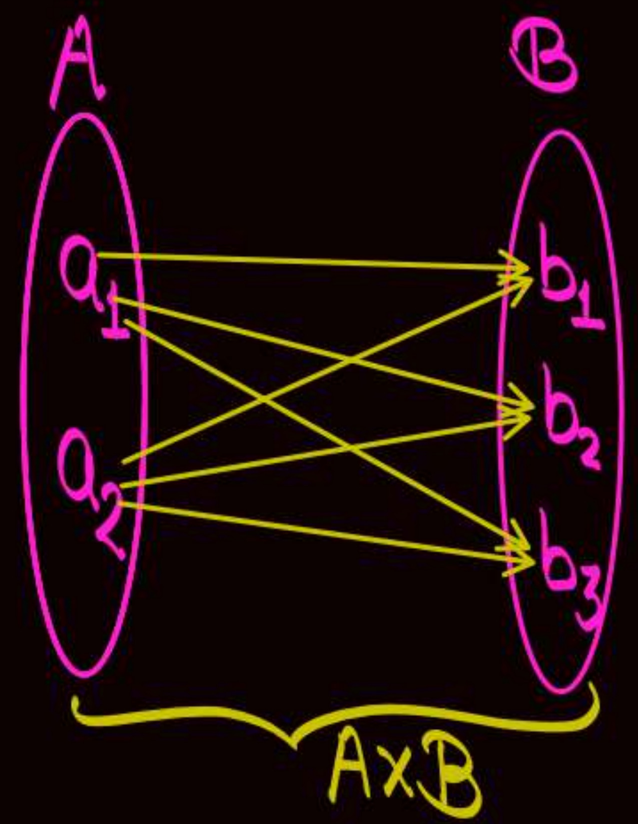
* If any of A or B is an Empty set, then $A \times B = B \times A = \{\ \} = \emptyset$

Note:-

$A = \{a_1, a_2\}$

$B = \{b_1, b_2, b_3\}$

$A \times B = \{(a_1, b_1), (a_1, b_2), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_2, b_3)\}$



In "$A \times B$" every element of set A relates with every element of set B.

$\therefore$ "$A \times B$" is the universal relation from set A to set B.

Note:

$$\boxed{\begin{array}{l} \text{If } |A| = m \text{ and } |B| = n \\ \text{then } |A \times B| = |A| \cdot |B| \\ \qquad |A \times B| = m \cdot n \end{array}}$$

- A relation from set A to set B defines that how exactly the elements of set A relates with elements of set B.

- In Cartesian product "$A \times B$", every element of set A relates with every element of set B.

  i.e, $A \times B$ is the universal relation from set A to set B

$\therefore$ Any relation from set A to set B is a subset of "$A \times B$" {i.e; Any relation is a subset of universal relation

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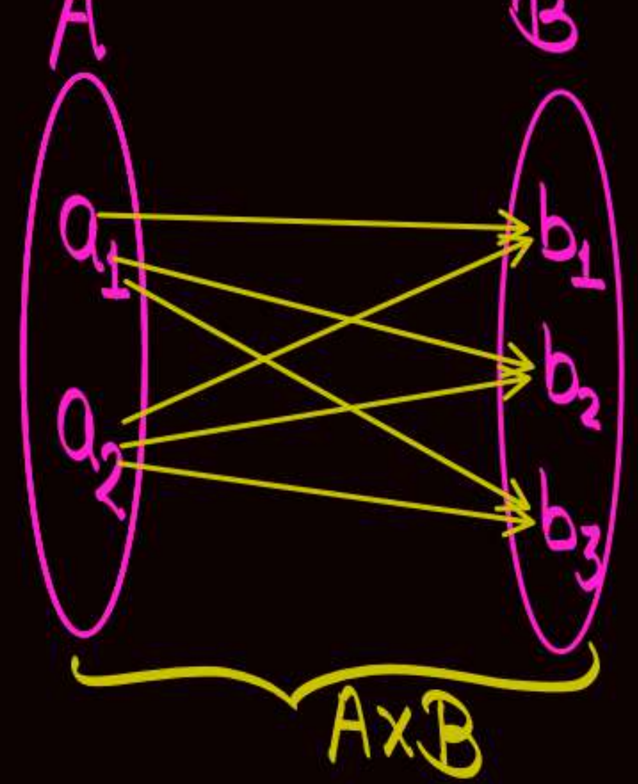
eg:-

$A = \{a_1, a_2\}$

$B = \{b_1, b_2, b_3\}$

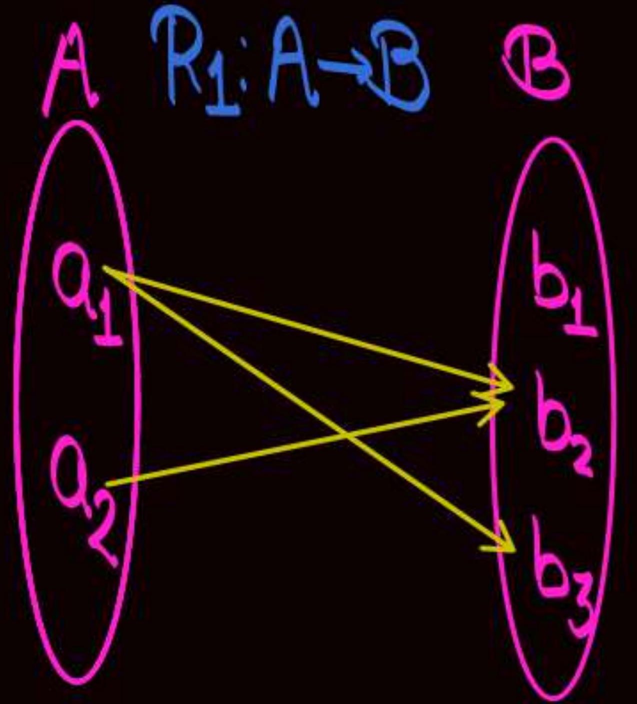
$A \times B = \{(a_1, b_1), (a_1, b_2), (a_1, b_3), (a_2, b_1), (a_2, b_2), (a_2, b_3)\}$



* Let R_1 is a relation from set A to set B, defined as

$$R_1 = \underbrace{\{(a_1, b_2), (a_1, b_3), (a_2, b_2)\}}$$

We can observe that R_1 is a subset of $A \times B$.



* Number of different relations possible from set A to set B

Let A and B are two sets such that $|A| = m$ and $|B| = n$, then

How many different relations are possible from set A to set B?

* Every relation from A to B is a subset of "$A \times B$"

$\therefore$ Number of different relations Possible from set A to set B = Number of Subsets of $A \times B$

$$= 2^{|A \times B|}$$

$$= 2^{|A| \cdot |B|}$$

$\therefore$ $\boxed{\text{Number of different relations Possible from set A to set B} = 2^{m \cdot n}}$

One Subset of "$A \times B$" is the Empty set, Empty Subset of $A \times B$ is also a relation from set A to set B, and it is called Empty relation

Topic : Relation

A relation from set A to set A it self is called a relation on set A

* Every relation on set A is a subset of "A x A"

* If $|A| = n$, then

Number of relations possible on set A = Number of Subsets of "A x A"

$= 2^{|A \times A|}$

$= 2^{(n^2)}$

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Topic : Types of Relations

All seven relations are always defined from set A to set A itself.
↓
i.e., on same set

(1) Diagonal Relation (Identity Relation)

(2) Reflexive Relation

(3) Irreflexive Relation

(4) Symmetric Relation

(5) Anti-symmetric Relation

(6) Asymmetric Relation

(7) Transitive Relation

(8) Complement of a relation

(9) Inverse of a relation

(10) Composite of two relations

Slide

Topic : Diagonal Relation / (Identity Relation)

A diagonal relation (Identity relation) on set A is denoted by Δ_A (I_A), and it is defined as.

it is the definition of a set

$$\Delta_A = \{ (a, a) \mid a \in A \}$$

e.g. Let $A = \{1, 2, 3\}$

$R_1 = \{(1,1), (2,2)\}$, $3 \in A$, but $(3,3) \notin R_1$, $\therefore$ R_1 is not a diagonal relation on set A.

$R_2 = \{(1,1), (1,2), (2,2), (3,3)\}$, order pair (1,2) can never belong to diagonal relation because $1 \neq 2$

$R_3 = \{(1,1), (2,2), (3,3)\}$ it is the diagonal relⁿ on set A.

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Let $A = \{1, 2, 3, 4, \ldots, n\}$

All the elements are on Principle diagonal, Together they define the diagonal Rel^n

$$A \times A = \left\{ \begin{array}{ccccc} (1,1), & (1,2), & (1,3), & \ldots & (1,n) \\ (2,1), & (2,2), & (2,3), & \ldots & (2,n) \\ (3,1), & (3,2), & (3,3), & \ldots & (3,n) \\ \vdots & \vdots & \vdots & & \vdots \\ (n,1), & (n,2), & (n,3) & \ldots & (n,n) \end{array} \right\}$$

Note:-

On any given set A, Exactly "one" diagonal relation is possible

Note:

$\boxed{\text{Cardinality of diagonal relation on set } A = |A|}$

Let $A = \{1, 2, 3, 4, \ldots, n\}$

These order pairs are called diagonal order pairs

All other order pairs are called non-diagonal order pairs

$$A \times A = \left\{ \begin{matrix} (1,1), & (1,2), & (1,3), & \ldots & (1,n) \\ (2,1), & (2,2), & (2,3), & \ldots & (2,n) \\ (3,1), & (3,2), & (3,3), & \ldots & (3,n) \\ \vdots & \vdots & \vdots & & \vdots \\ (n,1), & (n,2), & (n,3) & \ldots & (n,n) \end{matrix} \right\}$$

If $|A| = n$, then

① Total no. of order pairs in $A \times A = n^2$

② Total number of diagonal order pairs in '$A \times A$' $= n$

& ③ Total number of non-diagonal order pairs in $A \times A = (n^2 - n)$

Topic : Reflexive Relation

- A relation 'R' on set A is called a reflexive relation if and only if,

$$a \overset{R}{\ } a, \ \forall a \in A$$

i.e. $(a, a) \in R, \ \forall a \in A$

it is the Constraint on the relation for it to be called a "reflexive relation"

"$a \overset{R}{\ } b$" is read as, 'a' relates with 'b' w.r.t. relation R.

i.e. $(a, b) \in R$

$a \overset{\not R}{\ } b$: It is read as 'a' does not relate with 'b'.

i.e. $(a, b) \notin R$

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eg: Let $A = \{1, 2, 3\}$

① $R_1 = \{(1,1), (2,2), (3,3)\}$ → "R_1 is a diagonal relation on set A as well as a reflexive relation on set A."

② $R_2 = \{(1,1), (2,2), (3,1) (3,3)\}$ "R_2" is not a diagonal relation on set A, but "R_2" is a reflexive relation on set A.

③ $R_3 = \{(1,1), (2,2)\}$ "R_3" is neither a reflexive relation on set A nor a diagonal relation on set A.

④ $R_4 = \{(1,2), (2,3), (3,1)\}$ "R_4" is neither a reflexive relation on set A nor a diagonal relation on set A.

Note :-

1. Every diagonal relation on set A is also a reflexive relation on set A, but Every reflexive relation on set A need not be a diagonal relation on set A.

2. Diagonal relation on set A is the smallest reflexive relation on set A.

3. Largest reflexive relation on set A is "A x A"

Q: Let $A = \{1, 2, 3, \ldots, n\}$
How many reflexive relation are possible on set A
Where $|A| = n$.

Reflexive Relation = (All 'n' diagonal order pairs must be present) and (Along with all diagonal order pairs, any number of non-diagonal order pairs may be present)

Choose all 'n' out of 'n' diagonal order pairs

Choose any number of non-diagonal order pairs out of (n^2-n)

$\therefore$ Number of Reflexive Rel^n = $$\left({}^{n}C_{n}\right) * \left\{ {}^{n^2-n}C_0 + {}^{n^2-n}C_1 + {}^{n^2-n}C_2 + \cdots + {}^{(n^2-n)}C_{(n^2-n)} \right\}$$

$$= 1 * 2^{n^2-n}$$

$$\boxed{= 2^{n^2-n}}$$

Note: Another approach :—

Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many reflexive relⁿ are possible on set A ?

'1' choice for each diagonal order pair. i.e, it must be present

'2' choices for each non-diagonal order pair → they may be present (or) they may be absent

Number of Reflexive Relⁿ =

$$\left\{\begin{array}{cccccc}
(1,1), & (1,2), & (1,3), & \ldots & (1,n) \\
1 \times & 2 \times & 2 \times & - - \times & 2 \\
(2,1), & (2,2), & (2,3), & \ldots & (2,n) \\
2 \times & 1 \times & 2 \times & - - & \\
(3,1), & (3,2), & (3,3), & \ldots & (3,n) \\
2 \times & 2 & 1 & & 2 \\
\vdots & \vdots & \vdots & & \vdots \\
(n,1), & (n,2), & (n,3) & \ldots & (n,n) \\
2 \times & 2 \times & 2 & & 1
\end{array}\right\}$$

$$= \underbrace{1 \times 1 \times \ldots \times 1}_{n \text{ times '1'}} \times \underbrace{2 \times 2 \times 2 \times \ldots \times 2}_{(n^2-n) \text{ times '2'}} = 2^{n^2-n}$$

Ans

eg: Relation "$\le$" is a reflexive relation on any set of real numbers.

let $A = \{0, 1, 3\}$ is a set of real numbers

$\therefore$ $(a,b) \in \, \le$ if and only if "$a \le b$"

$$\le \; = \left\{ \begin{matrix} (0,0) & (0,1) & (0,3) \\ \cancel{(1,0)} & (1,1) & (1,3) \\ \cancel{(3,0)} & \cancel{(3,1)} & (3,3) \end{matrix} \right\}$$

Not present because $1 \nleq 0$

all diagonal order pairs are present.

$\therefore$ "$\le$" is reflexive on set of real numbers.

$\underbrace{}$ don't read as divided by.

eg: Relation divides ($\div$) is reflexive on any set of positive integer. { True, because Every positive integer divides it self }

eg: Relation subset ($\subseteq$) is reflexive on any set of sets { True, because Every set is a subset of itself }

~~eg: Relation divides ($\div$) is reflexive on any set of integers~~ { need not be true, because '0' may be present and '0' does not divide '0' }

Let $A = \{1, 2, 3, 4, \ldots, n\}$

$$A \times A = \left\{ \begin{matrix} (1,1), & (1,2), & (1,3), & \ldots & (1,n) \\ (2,1), & (2,2), & (2,3), & \ldots & (2,n) \\ (3,1), & (3,2), & (3,3), & \ldots & (3,n) \\ \vdots & \vdots & \vdots & & \vdots \\ (n,1), & (n,2), & (n,3) & \ldots & (n,n) \end{matrix} \right\}$$

2 mins Summary

Topic: Cartesian product

Topic: Relation

Topic: Different types of relations

Topic: Diagonal relation

Topic: Reflexive relation

Topic: Irreflexive relation

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THANK - YOU

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